

Same Storm, Different Boats: Generative AI and the Age Gradient Within Firms^{*}

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Abstract

From 2022 to 2026, Swedish job postings fell while share prices rose, coinciding with monetary tightening from April 2022 and ChatGPT's release that November. Aggregate postings show no clear link to either explanation, but advertising within employers shifts away from AI-exposed occupations after the release. Payroll records are clearer. From 2023 to 2024–25, the young-to-older employment ratio at ages 22 to 25 fell 3.9 per cent more at exposed employers, with separations rising and no significant change in hiring. The oldest workers gained most. Among the young, women's ratio fell 7.4 per cent and men's did not detectably change.

Keywords: Generative artificial intelligence, Job postings, Labour demand, Employment composition, Monetary policy

JEL: J23, J24, O33

1. Introduction

After years of zero and negative policy rates, central banks began raising them in 2022, in response to inflation that built during the recovery from the pandemic and rose further after Russia's invasion of Ukraine. The Riksbank's first rise came in April 2022. Seven months later, on 30 November

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2022, ChatGPT was released to the public. Swedish job postings then fell from their peaks in 2022 and 2023, while share prices kept rising (Figure 1). Generative AI and monetary tightening offer competing explanations. One reading is that firms hire fewer workers for the codified tasks young workers often perform (Brynjolfsson et al., 2026). The other is that tightening cooled hiring. A decline in young workers’ relative employment can reflect fewer jobs for the young, growth among older colleagues, or both. We ask whether, inside the employers most exposed to generative AI, the employment of young workers relative to their older colleagues changed as AI use spread, and whether their advertising shifted before or after the launch.

We address this question with two Swedish data sources: 4.9 million job advertisements from Platsbanken, the Public Employment Service’s job board, matched to the Dynamic AI Occupational Exposure (DAIOE) index (Engberg et al., 2024), which scores 96 per cent of them, and monthly payroll declarations covering every employee in Sweden. US payroll panels select employers and full-time workers, and comprehensive US flows come in industry-state-age cells. The Swedish declarations allow within-employer comparisons in every sector, including part-time work and the public sector, a third of employees (Statistics Sweden, 2026a). Within each employer and month, we compare young workers with older colleagues, differencing out shocks that affect employment proportionally across age groups. Shocks that disproportionately affect young workers, including potential credit responses to monetary tightening, remain and are examined in Section 3. Because the occupation register is published about two years late (Statistics Sweden, 2026c), classifying workers by their own codes can by itself produce a spurious decline among the young (Online Appendix IV). We therefore measure exposure at the employer, from the occupations its incumbents aged 31 to 69 had in the 2019 register, and follow each employer’s age composition in those declarations.

In short, the posting data cannot separate the two explanations, while inside exposed employers the employment of young workers falls relative to that of older colleagues after the launch, and the decline falls on young women. Across occupations, advertisements in highly exposed occupations fall relative to others with the rate rise on our exposure index and after the launch on the index of Eloundou et al. (2024), and neither date survives richer occupational controls. Within employers, advertising shifts away from exposed occupations after the launch. For employment, we compare January 2024 to June 2025 with the thirteen months from December 2022, the first year

after the launch, before firms’ reported AI use rose from 10 to 25 per cent. The young-to-older employment ratio at ages 22 to 25 fell 3.9 per cent more at highly exposed employers than at less exposed ones. The estimate holds under industry-by-age shocks, employer leverage, older comparison groups below every pension age, monthly seasonality, a continuous exposure score, a stricter size floor and industry clustering. It is as large against the months before the rate rise as against 2023, and the same design moved back two years returns zero (Section 3; Online Appendix III.2 and III.3). Older workers gain relative to those aged 41 to 49, increasingly with age from 50. On both exposure indices the shortfall falls on young women. The estimate for men and women together is smaller and insignificant on the Eloundou measure. Relative separations of the young rise, though imprecisely under industry clustering, while hiring shows no significant change.

A first strand of the literature finds early-career and junior employment falling relative to senior employment in AI-exposed jobs and firms (Brynjolfsson et al., 2026; Hosseini Maasoum and Lichtinger, 2026). Across countries, this mainly reflects growth in senior employment (Chandar and Klein Teeselink, 2026). Administrative and cross-country evidence is more mixed, and weighs monetary shocks, earlier trends and labour-market institutions as competing explanations (Tucker, 2026; Bonin et al., 2026; Humlum and Vestergaard, 2025; Kauhanen and Rouvinen, 2026; Facius and Iacono, 2026; Klein Teeselink and Carey, 2026). The Nordic register studies differ from ours in sample restrictions and sample sizes, and in exposure indices (Kauhanen and Rouvinen, 2026; Facius and Iacono, 2026; Humlum and Vestergaard, 2025; Bonin et al., 2026). Labour-market regulation differs too: of the four countries, Sweden combines the strictest protection of permanent contracts with the least regulated temporary contracts (OECD, 2026), which the young more often hold (Statistics Sweden, 2026a). Our declarations cover the whole workforce, including small employers, part-time work and the public sector, which allows within-employer comparisons on 104,217 employers (Online Appendix III.5).

We contribute to the early-career employment literature in three ways. First, comparing age groups inside employers in every sector, we find the decline of the young matched by relative gains of older workers and concentrated among young women. Second, employers’ advertising mix responds to the launch, while market-wide advertising cannot be dated to it. Third, re-estimating with the occupation codes available in each past year, we show that the register’s delay can by itself produce a large apparent early-career

decline, which scoring employers on their 2019 workforce avoids.

2. Data and empirical strategy

Job-posting margin. We use all advertisements published on Platsbanken from January 2020 through June 2026, after dropping duplicates and advertisements without an occupation field (Online Appendix II.7 and II.8); coverage drift towards other platforms would enter the posting estimates (Online Appendix VI.1).¹ Exposure is the DAIOE index (Engberg et al., 2024), the 2023 vintage of the generative-AI percentile of each four-digit occupation (SSYK 2012, Sweden’s version of ISCO-08), cut into quartiles with Q4 the most exposed. Cells are occupation-by-month. A within-employer analysis asks whether employers shift their advertising towards less exposed occupations (Online Appendix V).

The baseline specification is

$$\ln A_{it} = \alpha_i + \lambda_t + H_i(\beta_1 P_t^{\text{RB}} + \beta_2 P_t^{\text{GPT}}) + \varepsilon_{it}, \quad (1)$$

where A_{it} counts advertisements in four-digit occupation i and month t , α_i and λ_t are occupation and month fixed effects, P_t^{RB} and P_t^{GPT} are cumulative indicators from April 2022, the month the rate rise was announced, and from December 2022, the first full month after the launch, and $H_i = 1$ if occupation i is in the top quartile of generative-AI exposure. Standard errors are clustered at the occupation level. Cells with no postings, 2.4 per cent, drop out of the log. Poisson estimation on the balanced panel with zeros gives similar estimates (Online Appendix II).

Employment margin. We ask whether the young-to-older employment ratio changes more at highly exposed employers than at less exposed ones. Exposure is fixed by each employer’s pre-pandemic occupational composition, so later employment requires no current occupation code. The identifying assumption is that, without the exposure-related change under study, the relative age composition would have evolved similarly across exposure groups after accounting for seasonality. Employer shocks with age-specific effects

¹Bulk files (2020–2025) and closed-quarter files for 2026 from <https://data.jobtechdev.se/annonser/historiska/>; coverage in Online Appendix VI.1 (Cronert, 2022).

are the threat to it. Section 3 tests industry-specific movements and leverage. The estimates describe relative workforce composition, not aggregate employment effects.

Employment comes from monthly employer declarations at the individual level (*Arbetsgivardeklaration på individnivå*) from Statistics Sweden, which cover every employee in Sweden, linked to birth year and sex from the annual LISA register. Any declaration record counts as a person-month. Counting only records with cash pay barely moves the estimate (Online Appendix VI.1). An employer is scored by the occupations its own incumbents aged 31 to 69 had in the 2019 register: the employment-weighted mean of their DAIOE percentiles, each occupation scored at the three-digit level (Online Appendix IV.4), cut into quartiles. The scored employers held 98 per cent of 2019 payslips and by 2025 scored employers hold 92 per cent of person-months. Employers founded after 2019 cannot be scored, so the design does not capture employment changes through firm entry (Online Appendix VI.1). The floor at age 31 keeps the young out of their own exposure measure and puts the score on the codes the register keeps most current (Online Appendix IV.1). In Statistics Sweden’s 2023 survey, top-quartile firms are 20.9 percentage points likelier to report using AI. The score also predicts AI use from before ChatGPT, so we compare within the employer and do not treat exposure as exogenous (Online Appendix III.2). The panel runs from January 2021 to June 2025.

For each young band we estimate a count specification that identifies changes in its employment relative to older colleagues within employers:

$$\mathbb{E}[N_{fat}] = \exp\left(\alpha_{fa} + \delta_{ft} + \lambda_{at} + H_f Y_a [b_R D_t^R + b_I D_t^I + b_L D_t^L + \sum_{s=1}^3 \theta_s S_{st}]\right), \quad (2)$$

where f indexes employers, a age bands and t months, N_{fat} counts workers of band a at employer f in month t , H_f marks the top quartile of the fixed employer score, built on its 2019 workforce, and Y_a the young band, aged 22–25 or, in a separate fit, 26–30. The panel holds the young band and the four bands aged 31 to 69. The indicators D_t^R , D_t^I and D_t^L mark three mutually exclusive periods, the tightening months (April to November 2022), the interim period (December 2022 to December 2023) and the later period (January 2024 to June 2025), each read against January 2021 to March 2022, and S_{st} are calendar-quarter indicators, the fourth quarter omitted. The

headline is $\tau = b_L - b_I$, the log relative change in the young-to-older ratio from the interim to the later period at top-quartile relative to less exposed employers; $100[\exp(\tau) - 1]$ is the percentage change, and $\hat{\tau} = \hat{b}_L - \hat{b}_I$ its estimate. Online Appendix III.2 reports the same model with cumulative period indicators. For the age profile we extend Equation (2) to eight bands, with band-specific period coefficients against those aged 41–49 (Figure 2).

The effects α_{fa} , δ_{ft} and λ_{at} absorb an employer’s persistent age structure, its proportional shocks and national age shocks, and the calendar terms remove a seasonal cycle in the ratio that predates the launch; month-of-year terms in their place leave $\hat{\tau}$ unchanged (Online Appendix III.2). Differencing two ages inside one employer also removes the firm-wide scale effect of adoption (Chandar and Klein Teeselink, 2026). We estimate by Poisson pseudo-maximum likelihood on a balanced, zero-filled panel, because zero cells are common at these ages and a log-plus-one coefficient is not a percentage change when employment can be zero (Chen and Roth, 2024). Standard errors are clustered by employer. A hire is an employer-person pair present in a month and absent from that employer the month before, and a separation the reverse.

3. Results

Figure 1 shows the Swedish divergence: the OMX Stockholm 30 rose by more than two thirds from February 2020 to mid-2026, while postings declined from their peaks, April 2022 in the most exposed quartile and April 2023 in the others. Relative to the other three quartiles, postings in top-quartile occupations fell by about 12 per cent with the rate rise. After the launch of ChatGPT the point estimate implies a further decline of about 6 per cent, with a 95 per cent confidence interval that includes zero (Online Appendix II). Both estimates change little under Poisson estimation or with one-digit occupation group by month-of-year effects, and exposure is not detectably correlated with revealed rate sensitivity (Online Appendix II.2, II.4, II.6). The date with which the decline coincides depends on the exposure ranking, however. On the measure of Eloundou et al. (2024) it falls after the launch rather than with the rate rise, and on either measure neither date survives a linear trend interacted with exposure or one-digit group by year-month effects (Online Appendix II.2).

Each advertisement records its employer, so the comparison can also be drawn within the employer, using no register. Within employers, the rate-rise

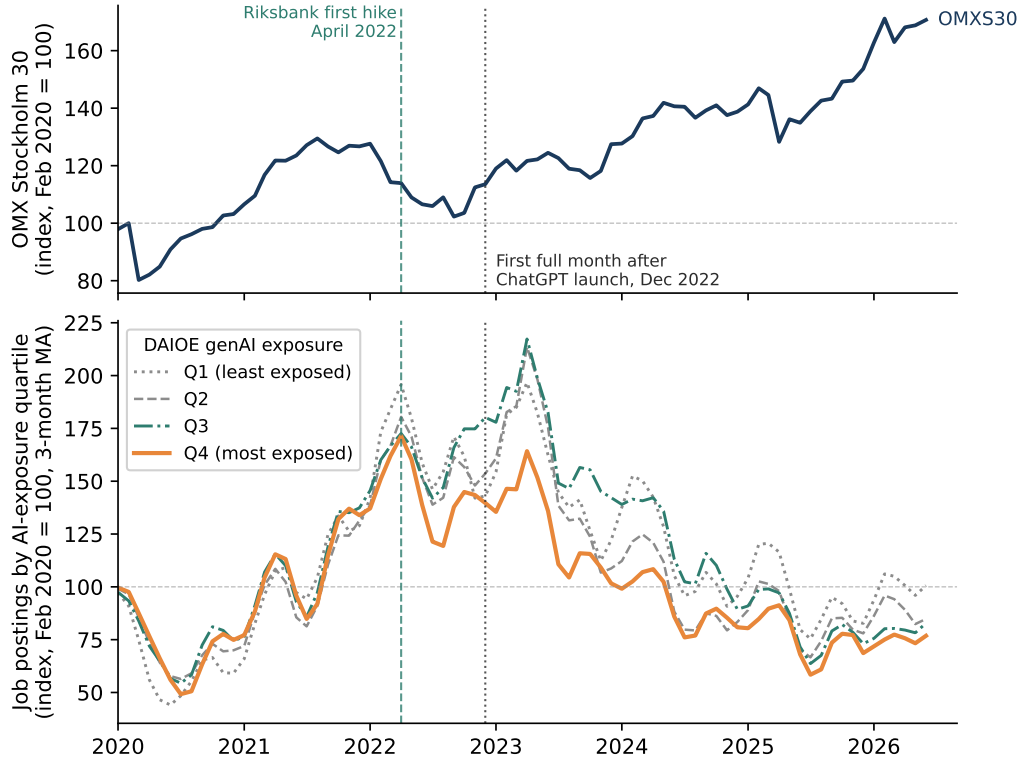


Figure 1: Stock market and job postings by AI-exposure quartile, Sweden 2020 to 2026. Upper panel: the OMX Stockholm 30, monthly averages of daily closes, February 2020 = 100. Lower panel: Platsbanken postings by generative-AI exposure quartile on the same base, three-month centred moving average. Vertical lines mark the first Riksbank rate rise (April 2022) and the first full month after the ChatGPT launch (December 2022).

interaction is indistinguishable from zero, while after ChatGPT an employer's advertisements in top-quartile occupations fall about 15 per cent relative to its other advertisements (Online Appendix V). Advertisements that ask for no experience fall too, but not distinguishably more than the rest.

The employment result compares the young with their older colleagues. Comparing January 2024 to June 2025 with the thirteen months from December 2022, the young-to-older employment ratio at ages 22 to 25 fell 3.9 per cent more at highly exposed employers than at less exposed ones ($\hat{\tau} = -0.040$, SE 0.010; Table 1). The same contrast at ages 26 to 30 is also 3.9 per cent.

Which of those colleagues gain? Figure 2 reports every band against the prime-aged on the same contrast. The young also fall relative to the

prime-aged, by less than relative to all older colleagues, and the relative gain rises with age, to 8.9 per cent at 65 to 69. The raw totals have the same shape: young headcount fell at exposed and less exposed employers alike, while the workforce aged 50 and over grew most in the top quartile. Hiring fell at every age below 50 in every quartile, at 22 to 25 by about as much as at 41 to 49. Because the young were hired at two to four times the monthly rate before 2024, an equal fall in hiring reduced their employment more (Online Appendix III.1). This common decline is differenced out of our estimates. In the United States, studies that follow entry rather than employment stocks find tentative evidence of fewer job starts among workers aged 22 to 25 in exposed occupations (Massenkoff and McCrory, 2026) and lower initial employment among bachelor’s graduates of the most exposed majors (Orr et al., 2026). In Sweden, the early-career earnings of university graduates have fallen more since 2023 the more exposed their field of study, and a falling share of recent graduates works at the most exposed firms (Nordström Skans and Sokolow Romin, 2026). On a check that scores each band from its own occupations, the oldest band’s gain is linked as much to teleworkability as to AI exposure, so remote work, which raises the cost of supervising and developing juniors (Lambert and Schindler, 2026), is an alternative reading for that band (Online Appendix II.6, III.2). For the youngest band the exposure estimate holds when employer teleworkability enters the same fit (-0.053 , SE 0.018), while teleworkability itself has no detectable effect (Online Appendix III.2). The gap between young women and young men is drawn inside the same employers, so a gain common to both sexes cancels from it (Table 1).

The relative decline falls on young women. On the same contrast, young women’s ratio to their older colleagues falls, while young men’s shows no detectable change. The female differential, a triple difference inside the same firms, is 6.9 per cent (-0.071 , SE 0.011; Table 1). US payroll data likewise show steeper declines for young women, which Brynjolfsson et al. (2026) attribute to their concentration in exposed occupations. Ours is drawn within employers. It shows no detectable linear drift and keeps two thirds of its size under industry by age by sex by month effects (Online Appendix III.2). Within broad fields of education, which Swedish evidence links to AI exposure (Nordström Skans and Sokolow Romin, 2026), the differential retains most of its pooled size (Online Appendix III.2). The design does not identify why young women fare relatively worse. Three readings fit a gap inside the employer: young women more often hold exposed occupations, a gap

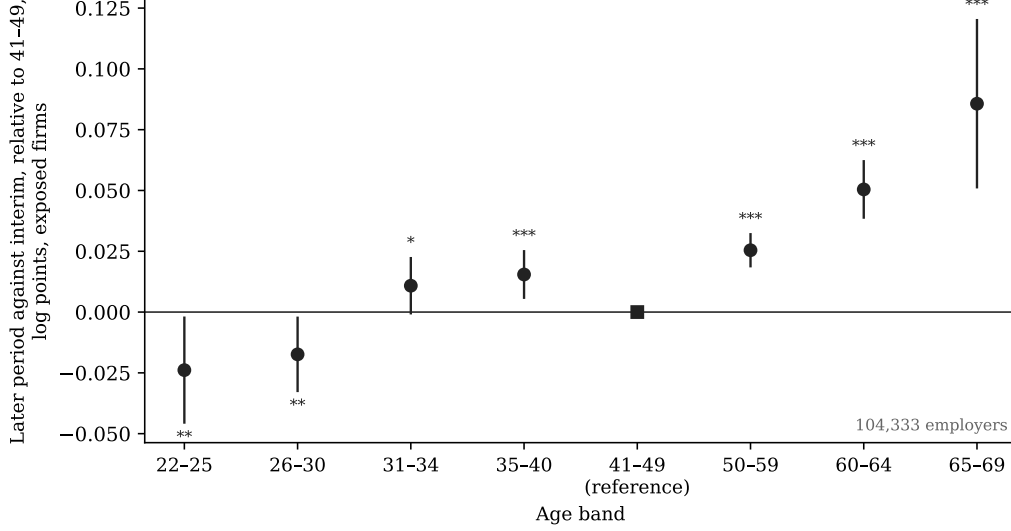


Figure 2: Exposure-related changes in the age profile: each band against ages 41–49, top-quartile against less exposed employers, the later period against the interim period ($\hat{\tau}$), with the calendar-quarter terms. Poisson on one panel of eight bands, 104,333 employers; exposure from the employer’s 2019 workforce. Whiskers: 95 per cent intervals clustered by employer; stars: p below 0.10, 0.05 and 0.01.

Table 1: Exposure-related changes in within-employer employment composition

January 2024 to June 2025 relative to December 2022 to December 2023

	Estimate	SE, employer	SE, industry
Ages 22–25 relative to older colleagues	−0.040	(0.010)	(0.019)
Young women, 22–25, relative to older colleagues	−0.077	(0.011)	(0.018)
Young men, 22–25, relative to older colleagues	−0.006	(0.012)	(0.023)
Young women minus young men ($\hat{\tau}_F - \hat{\tau}_M$)	−0.071	(0.011)	(0.014)
Ages 26–30 relative to older colleagues	−0.040	(0.007)	(0.014)

Poisson estimates of $\tau = b_L - b_I$ in Equation (2). Every row compares top-quartile with less exposed employers on the same contrast, the later period against the interim period. Older colleagues are those aged 31 to 69 at the same employer. The sex rows come from one specification that interacts every treatment term with being female, with sex-specific employer-by-age and age-by-month effects; their older colleagues are of both sexes. Standard errors are clustered by employer and by three-digit industry. In per cent, a row is $100[\exp(\tau) - 1]$. Employers: 104,217 at 22–25 and 117,090 at 26–30.

nationally widest at these ages (Online Appendix III.2); fixed-term contracts, whose non-renewal registers as a separation (Statistics Sweden, 2026a); and a move of young women from work into study, which payroll records do not observe. In 2024 they also used generative AI less often than young men (Statistics Sweden, 2025).

Figure 3 shows how the gap evolves from the launch quarter: the ratio of young to older employment at exposed employers, which had risen during the tightening months (Online Appendix III.2), turns down and declines through 2023 and 2024. The movement is gradual, not a break. We place the boundary between the periods at January 2024, when Statistics Sweden’s annual survey records AI use among enterprises with at least ten employees rising from 10 to 25 per cent (Statistics Sweden, 2026b, 2024). The date is when the survey measured the rise, not a break in the data. Moving it anywhere in the year around it leaves the estimate unchanged. Nor does the estimate rest on 2023 as its base: against the months before the rate rise, the ratio in the later period stands 4.1 per cent lower (-0.042 , SE 0.019), and over 2023 it was near that earlier level (-0.002 , SE 0.014; Online Appendix III.2). Timing alone does not exclude delayed monetary or other demand effects, but the shortfall persists through more than a year of easing.

For the youngest band the pre-period shows no detectable drift. In the two years before the launch its relative trend is half a standard error from zero, while that of the older young band rises, so that band is read against a rising pre-trend. The ratio did move in the pandemic year, by as much as in 2024 to 2025: a design backdated to that year picks it up (-0.038 , SE 0.008), one backdated to 2021 returns zero ($+0.004$, SE 0.007). That movement fell on young men: backdated to the pandemic year, the female differential is positive ($+0.052$, SE 0.010), the opposite of its sign since 2024 (Online Appendix III.2). Absorbing three-digit industry by age and month retains most of $\hat{\tau}$ for the youngest band and a quarter of it for the older one. Allowing the change to differ by employer leverage, the credit channel of tightening, leaves $\hat{\tau}$ unchanged (Online Appendix III.2). Clustered by industry, $\hat{\tau}$ remains significant at the five per cent level (Table 1).

Two national policy changes fall in the window: the 2020 and 2023 increases in pension ages, and the end of the reduced payroll contribution for young workers in March 2023. Month-by-age effects absorb their common effect, so either enters the estimate only through an effect on the young-to-older ratio that differs with exposure. Against colleagues below every pension age, $\hat{\tau}$ keeps most of its size (Online Appendix III.2).

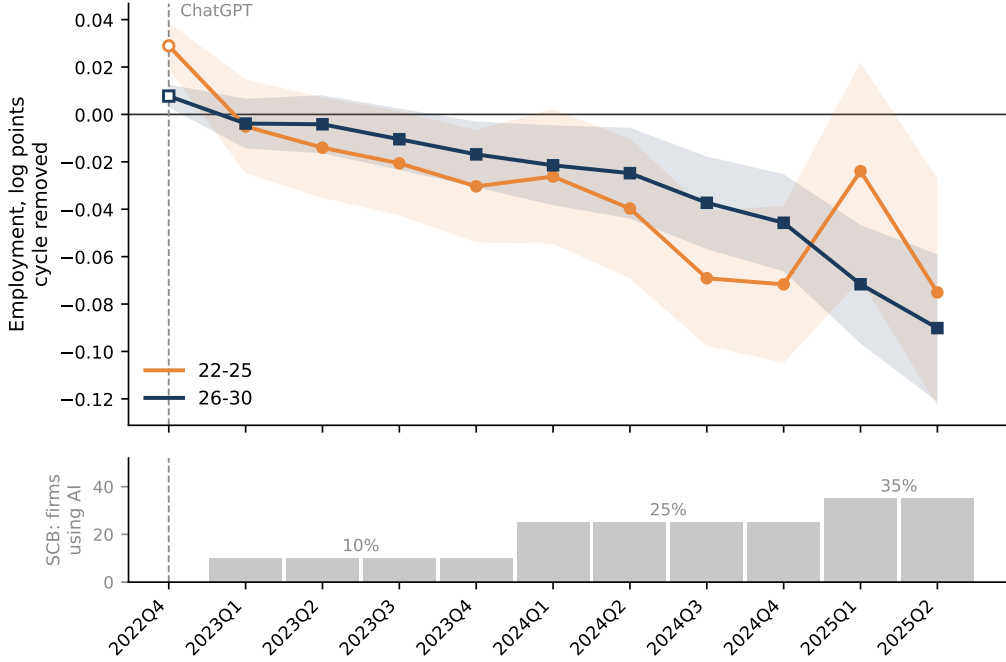


Figure 3: Quarterly exposure-related changes in young-to-older employment: top-quartile against less exposed employers, each quarter against the tightening months (April to November 2022), calendar cycle removed; the hollow 2022Q4 point is December 2022 alone. Shaded bands: 95 per cent intervals clustered by employer. Lower panel: Statistics Sweden's annual share of firms using AI, measured each January and drawn as a step.

On the headline contrast, relative separations of the youngest rise by 3.0 per cent, imprecisely once clustered by industry, while hiring shows no significant change (Online Appendix III.2). In 2023 more than half of employees under 25 held fixed-term contracts (Statistics Sweden, 2026a), whose non-renewal registers as a separation. The declarations record no contract type. A correction of hiring done during the tightening would also raise separations, although the ratio ends below its level before the rate rise (Online Appendix III.2). Among Danish adopters, whose decline is carried by firms with fewer than 100 full-time equivalents, the adjustment runs through hiring instead (Bonin et al., 2026), a difference that employment protection or pre-launch hiring may explain.

An independent reconstruction from the raw payroll declarations, without occupation fields, returns identical employment counts and the same estimate

(Online Appendix VI.1). Continuous exposure gives a decline of about one per cent per standard deviation. Quartile estimates locate it among the most exposed employers (Online Appendix III.2). The decline is smaller and imprecise among employers scored from very few incumbents (Online Appendix III.3). Classifying employers by the rating of Eloundou et al. (2024) leaves the female differential and the estimate for the older young band almost unchanged. For the youngest it gives -0.024 (SE 0.019), against -0.039 (SE 0.010) on DAIOE for the same employers: smaller and less precise. Where the two rankings disagree, DAIOE’s is the more closely associated with reported AI use (Online Appendix III.2).

4. Conclusion

As generative AI spread, the Swedish employers most exposed to it tilted towards older workers. From 2023 to 2024–25, the ratio of workers aged 22 to 25 to their older colleagues fell 3.9 per cent more at exposed employers than at less exposed ones, and the decline fell on young women. Job advertisements cannot separate monetary tightening from generative AI, although employers moved their advertising away from exposed occupations after the launch of ChatGPT. Payroll records go further. At exposed employers, the ratio rose in the first months of rate rises and turned down only after the launch, and the shortfall persisted through more than a year of easing. Moreover, the estimate is unchanged when we allow it to vary with employer leverage, the channel through which tightening reaches firms, and it holds when we control for employer teleworkability. The payroll records capture the age composition of existing employers, not aggregate employment. However, these employers account for 92 per cent of employment, so the shift matters for where most young workers begin their careers and gain experience.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic), models of the Opus, Sonnet and Fable families, for code, data processing,

editing and analytical review, and ChatGPT (OpenAI) for manuscript review and analytical discussion. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Data availability

Platsbanken job advertisements are publicly available under a CC0 licence (<https://data.jobtechdev.se/annonser/historiska/>). The employment analysis uses confidential microdata from Statistics Sweden, accessed via its MONA platform under the ethical approvals listed below (Online Appendix VI.1). Code is at <https://github.com/Magnus-L/canaries-sweden>; a DOI-bearing archive follows on acceptance.

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References

Bonin, S.M., Darougheh, S., Kuchler, A., 2026. AI Adoption and Firm Size: Employment Effects in Monthly Administrative Data. Working Paper 223. Danmarks Nationalbank. URL: <https://www.nationalbanken.dk/en/news-and-knowledge/publications-and-speeches/working-paper/2026/ai-adoption-and-firm-size-employment-effects-in-monthly-administrative-data>. 16 September 2026.

- Brynjolfsson, E., Chandar, B., Chen, R., 2026. Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence. Working Paper. Stanford Digital Economy Lab. URL: https://digitaleconomy.stanford.edu/app/uploads/2026/08/Canaries_August2026.pdf. August 2026 version; first version August 2025.
- Chandar, B., Klein Teeselink, B., 2026. How Does AI Change Labor Demand? Evidence from 41 Countries. Working Paper. Stanford University and King's College London. URL: <https://digitaleconomy.stanford.edu/app/uploads/2026/09/HowDoesAIChangeLaborDemand.pdf>, doi:10.2139/ssrn.7498743. 20 September 2026.
- Chen, J., Roth, J., 2024. Logs with zeros? Some problems and solutions. *The Quarterly Journal of Economics* 139, 891–936. doi:10.1093/qje/qjad054.
- Cronert, A., 2022. When the paper tiger bites: Evidence of compliance with unenforced regulation among employers in Sweden. *Regulation & Governance* 16, 1141–1159. doi:10.1111/rego.12383.
- Eloundou, T., Manning, S., Mishkin, P., Rock, D., 2024. GPTs are GPTs: Labor market impact potential of LLMs. *Science* 384, 1306–1308. doi:10.1126/science.adj0998.
- Engberg, E., Görg, H., Lodefalk, M., Javed, F., Långkvist, M., Monteiro, N.P., Kyvik Nordås, H., Pulito, G., Schroeder, S., Tang, A., 2024. AI Unboxed and Jobs: A Novel Measure and Firm-Level Evidence from Three Countries. Discussion Paper 16717. IZA.
- Facius, D., Iacono, R., 2026. Labor Market Consequences of Generative AI: Early Evidence from Norway. Working Paper 12752. CESifo. URL: <https://ssrn.com/abstract=6956666>, doi:10.65864/8a4auliirt.
- Hosseini Maasoum, S.M., Lichtinger, G., 2026. Generative AI as Seniority-Biased Technological Change: Evidence from U.S. Résumé and Job Posting Data. Working Paper. Harvard University. doi:10.2139/ssrn.5425555. SSRN 5425555, version of 6 June 2026.
- Humlum, A., Vestergaard, E., 2025. Still Waters, Rapid Currents: Early Labor Market Transformation under Generative AI. Working Paper 33777. National Bureau of Economic Research. doi:10.3386/w33777. Revised

- March 2026. Previously circulated as “Large Language Models, Small Labor Market Effects”.
- Kauhanen, A., Rouvinen, P., 2026. AI Has Not Impacted the Youth Labor Market in Finland. Working Paper 135. ETLA Economic Research.
- Klein Teeselink, B., Carey, D., 2026. AI, Automation, and Expertise. Working Paper SSRN 6134506. King’s College London and AI Objectives Institute. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6134506, doi:10.2139/ssrn.6134506. Version of 17 April 2026.
- Lambert, P.J., Schindler, Y., 2026. The Broken Ladder: AI, Remote Work, and Early-Career Hiring. Working Paper. SSRN. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6787638, doi:10.2139/ssrn.6787638. SSRN Working Paper 6787638, May 2026.
- Massenkoff, M., McCrory, P., 2026. Labor market impacts of AI: A new measure and early evidence. Anthropic Research, 5 March 2026. URL: <https://www.anthropic.com/research/labor-market-impacts>.
- Nordström Skans, O., Sokolow Romin, L., 2026. Structural Change and AI-Exposed Skills: Evidence from Swedish Real-Time Administrative Data. Working Paper. Uppsala University. URL: <https://drive.google.com/file/d/1DHVJWQg04xaA3dSi0yJ3Yi7308kigHNU/view>. version of 29 September 2026.
- OECD, 2026. OECD indicators of employment protection: Strictness of employment protection. OECD Data Explorer, dataflow OECD.ELS.JAI/DSD_EPL@DF_EPL, version 4. URL: <https://www.oecd.org/employment/emp/oecdindicatorsofemploymentprotection.htm>. individual dismissals (regular contracts) and temporary contracts, regulation in force on 1 January, 2015–2025; accessed 27 September 2026.
- Orr, C., Tucker, L.C., Warren, L., 2026. Graduating into Disruption: Labor Market Outcomes for AI-Exposed College Majors. Working Paper CES 26-56. U.S. Census Bureau, Center for Economic Studies. URL: <https://www.census.gov/library/working-papers/2026/adrm/CES-WP-26-56.html>.
- Statistics Sweden, 2024. Kvalitetsdeklaration: IT-användning i företag 2024. Quality declaration NV0116. Statistiska centralbyrån. URL: <https://www.>

scb.se/contentassets/9cfc4627e5954d109ab39dda9df1ba69/nv0116_kd_2024_.pdf. Section 1.2.5: the reference time for the target variables is January 2024, with exceptions for e-commerce, parts of the IT-specialist block and information security. The artificial-intelligence module is not among the exceptions.

Statistics Sweden, 2025. Användning av generativa AI-verktyg efter användning av internet, kön, redovisningsgrupp, tabellinnehåll och år. Statistikdatabasen, table LE0108T82, survey on ICT usage by individuals (LE0108). URL: https://www.statistikdatabasen.scb.se/pxweb/sv/ssd/START_LE_LE0108_LE0108Q/LE0108T82/. Used generative AI tools, 2024, ages 16–24: men 62, women 47 per cent; employed: men 33, women 22 per cent; accessed 27 September 2026.

Statistics Sweden, 2026a. Anställda 15–74 år (AKU) efter anknytningsgrad till arbetsmarknaden, sektor, typ av data, kön och ålder. år 2001–2025. Labour Force Surveys (AKU), Statistikdatabasen, table AKURLAnstSektorAr (TAB6525). URL: https://www.statistikdatabasen.scb.se/pxweb/sv/ssd/START_AM_AM0401_AM0401J/AKURLAnstSektorAr/. Fixed-term share computed as one minus permanent employees over all employees, all sectors; public-sector share computed as state plus municipal and regional employees over all sectors (33.6 per cent in 2025); accessed 25 September 2026.

Statistics Sweden, 2026b. Användning av AI-tekniker efter typ av teknik, redovisningsgrupp, tabellinnehåll och år: användning av minst en AI-teknik, företag med 10 eller fler anställda. Statistikdatabasen, table AiTeknikerTypN, survey on ICT usage in enterprises (NV0116). URL: https://www.statistikdatabasen.scb.se/pxweb/sv/ssd/START_NV_NV0116_NV0116M/AiTeknikerTypN/. Shares 2023: 10, 2024: 25, 2025: 35 per cent; accessed 22 September 2026.

Statistics Sweden, 2026c. Yrkesregistret med yrkesstatistik, 2024. Statistiknyhet, 16 January 2026. URL: <https://www.scb.se/hitta-statistik/statistik-efter-amne/arbetsmarknad/utbud-av-arbetskraft/yrkesregistret-med-yrkesstatistik/pong/statistiknyhet/yrkesregistret-med-yrkesstatistik-2024/>. employment measured in November of the reference year; accessed 28 September 2026.

Tucker, L.C., 2026. You're (not) Hired: Artificial Intelligence and Early Career Hiring in the Quarterly Workforce Indicators. Working Paper CES 26-27. U.S. Census Bureau, Center for Economic Studies. URL: <https://www.census.gov/library/working-papers/2026/adrm/CES-WP-26-27.html>.